



SLIDE 1 — Title

Pediatric vs Adult Dengue Severity Prediction with Subgroup-Specific SHAP Explanations

“Assalamu Alaikum and good morning everyone.”

“I am Mohammad Mazharul Alam, Roll Number 22070116, and my co-researcher is Md. Aftabul Islam, Roll Number 22070103.”

“Our thesis is titled ‘Pediatric vs Adult Dengue Severity Prediction with Subgroup-Specific SHAP Explanations.’”

“This work was conducted under the guidance of Mohammad Arif Hasan Chowdhury, Assistant Professor, Department of Computer Science and Engineering, University of Science and Technology Chittagong.”

“Today, I will present the background, problem statement, methodology, experimental results, SHAP-based explanations, web application, and future work of our study.”



SLIDE 2 — Overview

“This presentation is organized into several parts.”

“First, I will discuss the background and introduction. Then I will explain our objectives and problem statement, followed by related works.”

“After that, I will present our proposed methodology, experimental results and discussion, and finally our conclusion and future work.”



SLIDE 3 — Background

“Dengue severity prediction is an important problem because patients of different age groups may show different clinical patterns.”

“However, many machine learning models are trained using pooled data, where pediatric and adult patients are analyzed together.”

“When all age groups are combined into one model, important differences between children and adults may be hidden.”

“Therefore, our study focuses on comparing age-stratified models with a pooled baseline and investigating whether their explanations also differ.”



SLIDE 4 — Why this research?

“There are three main points behind our research.”

“First, pediatric and adult dengue patients can have different physiological characteristics, including differences related to plasma leakage, comorbidities, and hemorrhagic patterns.”

“Second, most existing machine learning studies use a single pooled model and often provide only one global SHAP explanation.”

“So, subgroup-specific differences may not be visible.”

“Third, our work combines age-stratified machine learning models, subgroup-specific SHAP analysis, and an early-stage risk-assessment dashboard.”



SLIDE 5 — Why this matters?

“Dengue is a mosquito-borne viral infection transmitted mainly by Aedes mosquitoes and remains an important public health problem in tropical and subtropical regions.”

“Early identification of patients who may progress to severe disease is important for timely clinical management.”

“A key limitation in previous machine learning approaches is the pooled-model assumption, where age is treated only as another input feature rather than as a stratification variable.”

“Our thesis therefore compares pooled and age-stratified models directly and also compares their SHAP explanations.”



SLIDE 6 — Problem Statement

“The main problem is that most classifiers are trained using pooled pediatric and adult data, assuming that one feature-severity relationship is suitable for all ages.”

“Also, when SHAP explanations are calculated globally, subgroup-level differences can be averaged out.”

“This can be different from clinical practice, because clinicians often consider age-specific reference patterns when evaluating patients.”

“Another problem is that many trained models are not connected to an explainable system that supports patient-level prediction and structured data collection.”

“Therefore, our research question is: Do subgroup-trained models and their SHAP explanations differ meaningfully from an age-pooled baseline, and can these differences be made usable through a dashboard?”



SLIDE 7 — Objectives

“Our study has three main objectives.”

“First, we develop pediatric, adult, and pooled machine learning models for severe-dengue prediction.”

“Second, we generate subgroup-specific SHAP explanations and identify model-derived feature splits.”

“Third, we develop an explainable web dashboard that can support longitudinal data collection.”



SLIDE 8 — Related Works

“Now I will briefly discuss the related works.”

“The first study used decision-tree algorithms for early-phase dengue diagnosis and outcome prediction. It considered age-related information, but it did not provide subgroup-specific SHAP explainability.”

“The second study focused specifically on pediatric dengue severity using early laboratory indicators. However, it did not provide a direct pediatric versus adult comparison.”

“The third study used clustering methods to discover clinical patterns. Age was identified as an important discriminator, but the study did not develop subgroup-specific prediction models.”



SLIDE 9 — Related Works

“The next group of studies used machine learning and explainability techniques.”

“One study used ANN, Random Forest, GBM, SVM, and Logistic Regression with SHAP for dengue severity prediction, but the analysis was based on a pooled cohort with global SHAP.”

“Another study combined stacking ensemble methods with SHAP and SMOTE, but again used global explanations.”

“A Bangladesh-based study used stacked ensembles with SHAP and LIME and reported high accuracy, but the explanation was still global rather than age-conditioned.”



SLIDE 10 — Research Gap

“The remaining studies also demonstrate the growing use of SHAP and machine learning for dengue analysis.”

“However, these studies mainly provide global explanations.”

“For example, SHAP-based factor analysis and hematology-driven machine learning studies did not specifically compare SHAP feature attribution between pediatric and adult groups.”

“Therefore, the main gap we identified is the lack of a direct quantitative comparison of age-specific model behavior and SHAP explanations.”

“Our work addresses this gap by training separate pediatric and adult models, comparing them with a pooled model, and analyzing their SHAP explanations.”

SLIDE 11 — Proposed Methodology / Workflow

“This slide shows the overall workflow of our proposed methodology.”

“We first collect and prepare the dengue dataset. Then we divide the data into pediatric, adult, and pooled cohorts.”

“For each cohort, we perform preprocessing and model training using five different classifiers.”

“After evaluating the models using multiple performance metrics, we select the appropriate models for further explainability analysis.”

“For the selected tree-based model, we apply SHAP TreeExplainer to understand feature contributions.”

“Finally, we extract model-derived feature splits and compare the SHAP feature rankings between pediatric and adult cohorts.”

SLIDE 12 — Data Preparation

“For our study, we used the Comprehensive Dengue Hematology and Clinical Dataset from Bangladesh.”

“From the dataset, we used nine core clinical and demographic predictors: WBC, HCT, RBC, lymphocyte percentage, neutrophil percentage, AST, ALT, age, and gender.”

“There were 1,329 raw records, from which 972 NS1-positive records were used in our analysis.”

“These records were divided into 158 pediatric patients, defined as age less than or equal to 18 years, and 814 adult patients, defined as age greater than 18 years.”

“For model development, we used an 80:20 stratified train-test split with random state 42.”

SLIDE 13 — Model & Evaluation

“For each cohort, we trained five classifiers.”

“These were Logistic Regression, Random Forest, XGBoost, LightGBM, and a Stacking Ensemble with Logistic Regression as the meta-learner.”

“We evaluated the models using six metrics: Accuracy, AUC-ROC, Sensitivity, Specificity, F1-score, and Matthews Correlation Coefficient, or MCC.”

“For explainability, we applied SHAP TreeExplainer to the best-performing model.”

“We then fitted depth-one decision trees, also called decision stumps, to SHAP-positive values to extract model-derived feature splits.”

“Finally, Kendall's tau-b was used to compare feature rankings between the pediatric and adult cohorts.”



SLIDE 14 — Dashboard Development

“We also developed a web-based system to make the trained models usable through an application.”

“The system uses JavaScript and Python.”

“ReactJS was used for the frontend, FastAPI was used for the backend, and PostgreSQL was used as the database.”

“The backend is connected with the cohort-specific best-performing models, allowing the system to provide predictions based on the appropriate model.”



SLIDE 15 — Model Performance Summary

“Now I will discuss the model performance.”

“For the pediatric cohort, Random Forest achieved the highest AUC of 0.961, with 90.6 percent accuracy, 66.7 percent sensitivity, 100 percent specificity, an F1-score of 0.800, and an MCC of 0.768.”

“The Stacking Classifier achieved the same accuracy, sensitivity, specificity, F1-score, and MCC, but its AUC was slightly lower at 0.947.”

“For the adult cohort, Random Forest performed best among the reported models, achieving an AUC of 0.892, accuracy of 84.7 percent, sensitivity of 78.1 percent, specificity of 90 percent, F1-score of 0.820, and MCC of 0.690.”

“The table also shows that XGBoost performed lower than Random Forest in both cohorts based on the reported metrics.”

SLIDE 16 — Cohort-Specific Selected Model

This slide **ସ୍ଥ** important.

“This slide summarizes the selected models for each cohort and compares them with the pooled baseline.”

“For the pediatric cohort, Stacking and Random Forest achieved the same accuracy of 90.6 percent. The Stacking model had an AUC of 0.947, while Random Forest achieved the higher AUC of 0.961.”

“For the adult cohort, Random Forest was the selected model with 84.7 percent accuracy and an AUC of 0.892.”

“For the pooled cohort, Random Forest achieved 79.0 percent accuracy, an AUC of 0.860, an F1-score of 0.739, and an MCC of 0.566.”

“Therefore, the age-stratified models performed better than the pooled baseline on the reported accuracy and AUC measures.”

 **If examiner asks: “Which model did you finally use for SHAP?”**

Say:

“We used the best-performing tree-based model for each cohort, because TreeSHAP is specifically suitable for tree-based models.”



SLIDE 17 — SHAP Feature Attribution

“After evaluating the models, we performed explainability analysis using SHAP.”

“SHAP, or SHapley Additive exPlanations, helps us understand how individual features contributed to model predictions.”

“This figure presents mean absolute SHAP values for the pooled, pediatric, and adult cohorts.”

“A larger mean absolute SHAP value indicates a larger average magnitude of contribution from that feature.”

“In the pooled cohort, Age and Lymphocyte percentage were among the most influential features.”

“In the pediatric cohort, Lymphocyte percentage was the most influential feature, followed by Neutrophil percentage and WBC.”

“In the adult cohort, Lymphocyte percentage was again the most influential feature, followed by Age and HCT.”

“Therefore, the relative importance of features changed between the pediatric and adult cohorts.”

“It is important to note that these are absolute SHAP values, so they show the magnitude of influence, not whether a feature increased or decreased the prediction.”



SLIDE 18 — Model-Derived SHAP Feature Splits

“After identifying influential features using SHAP, we performed model-derived split analysis.”

“For the pediatric cohort, the identified splits were Lymphocyte percentage greater than 44.50 percent, Neutrophil percentage at or below 48 percent, and WBC at or below 4.34 times 10 to the power of 3 per microliter.”

“For the adult cohort, the identified splits were Lymphocyte percentage greater than 24.99 percent, HCT greater than 40.65 percent, and Age greater than 31 years.”

“The important observation is that the same feature can have different model-derived split points across age groups.”

“For example, the Lymphocyte percentage split was 44.50 percent for the pediatric cohort, while it was 24.99 percent for the adult cohort.”

“This suggests that the model's decision behavior can differ according to age group.”

“However, these are model-derived thresholds, not validated clinical cutoffs.”

 এই শেষ lineটা must memorize.



SLIDE 19 — Web Application

“To make the research more practical, we developed a web application around the trained models.”

“The application provides a user interface where patient information can be entered and processed by the backend.”

“The backend uses the cohort-specific trained model to generate the prediction.”

“This allows the research model to move from an offline experimental environment toward an interactive risk-assessment system.”



SLIDE 20 — Web Application

“This slide shows another part of our web application.”

“The main purpose of the application is not only prediction, but also structured data collection and explainability.”

“By connecting the frontend, backend, database, and trained models, the system provides a practical interface for using the research framework.”

“The dashboard can also support future longitudinal data collection, which may be useful for developing temporal or sequential models in future work.”



SLIDE 21 — Conclusion & Future Work

এটা তোমার most important closing slide।

“To conclude, our results show that age-stratified models outperformed the pooled baseline on the reported performance measures.”

“For the pediatric cohort, the selected models achieved 90.6 percent accuracy, with an AUC of 0.947 for Stacking and 0.961 for Random Forest.”

“For the adult cohort, Random Forest achieved 84.7 percent accuracy and an AUC of 0.892.”

“Our SHAP analysis showed that the feature rankings were highly correlated between the pediatric and adult cohorts, with a Kendall's tau-b of 0.900.”

“However, despite this strong similarity in feature ranking, the model-derived feature-split thresholds differed between the age groups.”

“This suggests that a pooled model may hide some age-specific decision behavior.”

“For future work, we plan to perform external prospective validation using independent datasets from geographically different populations.”

“We also need a larger pediatric cohort to improve subgroup robustness.”

“Another important improvement is to use WHO-adjudicated severity labels instead of the current platelet-derived proxy.”

“Finally, longitudinal data collected through the dashboard could support future temporal or sequential modelling approaches, such as recurrent neural networks.”



SLIDE 22 — References

“These are the main references used to support our research.”

“They include previous studies on dengue diagnosis and severity prediction, machine learning approaches, stacking ensembles, and SHAP-based explainability.”

“We also used the Bangladesh dengue hematology and clinical dataset as the primary data source for our experiments.”

Don't read every reference aloud. **Just explain what the slide represents.**

SLIDE 23 — References

“These are the remaining references used in our study, including previous work on interpretable machine learning, SHAP-based dengue severity analysis, hematology-driven prediction, and optimized explainable machine learning.”

“These references helped us establish the research gap and position our age-stratified SHAP analysis within the existing literature.”

“Thank you everyone for listening.”

FINAL 30-SECOND CLOSING

যদি শেষে examiner বলে “Summarize your whole research in 30 seconds.”

এটা বলবে:

“In summary, our study compared pooled and age-stratified machine learning models for dengue severity prediction. We found that the pediatric and adult models performed better than the pooled baseline on the reported metrics. SHAP analysis showed that some important features were shared across age groups, but their model-derived split thresholds differed. Therefore, similar feature importance does not necessarily mean identical decision behavior. Finally, we integrated the cohort-specific models into a web-based dashboard for explainable risk assessment and future longitudinal data collection.”

তোমার পুরো thesis এক লাইনে

Presentation-এর আগে এই flow মাথায় রাখবে:

Problem→Age Stratification→ML Models→Evaluation→Best Models→SHAP→Feature Splits
→Kendall's Tau→Dashboard

আর তোমার central message:

“Similar feature rankings, but different feature-split behavior across age groups.”

এই sentence-টা তোমার পুরো thesis-এর **heart**।